

PISA 2025 Insight Report: Mongolia and Rwanda

This analysis compares PISA 2025 mathematics performance in Mongolia and Rwanda, with a pooled OECD comparison group as a broader benchmark. Mathematics is the primary outcome, while reading and science provide context. The analysis examines weighted achievement, socioeconomic status (ESCS), gender, grade repetition, school location, educational material shortages, and staff shortages. Results are descriptive rather than causal. PISA student weights and all 10 plausible values are used for achievement summaries, while correlations and subgroup comparisons describe associations with mathematics performance.

1. Package Imports

[Show code](#)[Show hidden output](#)

2. Downloading PISA Files

[Show code](#)[Show hidden output](#)

3. Data Loading and Column Selection

3A. Load School Data

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3B. Select and Load Student Data

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4. Column Standardization

4A. Standardize School Column Names

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Next steps:

[Generate code with school_df](#)[New interactive sheet](#)

4B. Standardize Student Column Names

[Show code](#)

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5. Data Cleaning & Missingness Analysis

5A. Define Missing-Value Codes

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5B. Map Missing Codes to Variables

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5C. Create Analysis Groups

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5D. Clean School Data and Check Missingness

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Next steps:

[Generate code with school_missing](#)[New interactive sheet](#)

5E. School Missingness Visualization

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5F. Clean Student Data and Check Missingness

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5G. Student Missingness Visualization

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Analysis Note: Data Missingness and Integrity

School identifiers are complete for Mongolia and Rwanda. Among OECD students, 3.19% lack a valid school identifier, which reduces coverage for analyses that attach school characteristics. Mongolia and Rwanda have at least 90% observed values for school location, educational material shortage, and staff shortage. OECD school-level missingness is higher: 12.04% for location, 13.67% for material shortage, and 13.59% for staff shortage.

Student identifiers, weights, gender, and all 30 plausible values are complete. Grade-repetition missingness is 0.50% in Mongolia, 5.06% in Rwanda, and 9.52% in the OECD comparison group. ESCS missingness is 3.43%, 6.67%, and 9.63%, respectively. Because students with missing ESCS later show lower mathematics scores than students with observed ESCS, socioeconomic comparisons may underrepresent some lower-performing students. Overall achievement estimates are unaffected by missing plausible values or weights, while school-condition and ESCS analyses use reduced samples.

6. Weighted Plausible Value Mean Estimation

For each subject and analysis group, the student-weighted mean is calculated separately for all 10 plausible values and then averaged across the 10 estimates.

6A. Weighted Plausible Value Function

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6B. Overall PISA Performance Summary

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```
=== PISA 2025 Weighted Performance Summary ===
```

	Mathematics	Reading	Science	
Rwanda	311.65	300.94	317.29	
Mongolia	418.64	374.25	424.03	
OECD	458.37	467.21	484.46	

Analysis Note: PISA Weighted Performance

Mongolia substantially outperforms Rwanda across all three domains. The mathematics gap is **106.99 points** (418.64 vs. 311.65), the science gap is **106.74 points** (424.03 vs. 317.29), and the reading gap is **73.31 points** (374.25 vs. 300.94). Both countries score highest in science and lowest in reading.

Mongolia also remains below the OECD comparison group by **39.73 points in mathematics**, **60.43 in science**, and **92.96 in reading**. Rwanda's gaps with the OECD comparison group are larger: **146.72 in mathematics**, **166.27 in reading**, and **167.17 in science**. Mathematics is the primary outcome for the remaining analysis.

7. Comparison of Student and School Characteristics

7A. Weighted Summary Functions

[Show code](#)

7B. Student and School Characteristics Summary

[Show code](#)

Analysis Group	Male Students (%)	Grade Repetition Rate (%)	Mean ESCS (Socioeconomic)	City or Larger School Location (%)	Educational Material Shortage (Mean)	Staff Shortage Index (Mean)
Mongolia	49.12	5.65	-0.78	47.06	0.04	0.06
Rwanda	43.16	55.70	-2.15	14.41	0.67	0.26
OECD	49.43	11.18	-0.19	48.94	-0.07	0.28

Analysis Note: Student and School Characteristics

Rwanda differs sharply from Mongolia in several student and school characteristics. Grade repetition is **55.70%** in Rwanda versus **5.65%** in Mongolia and **11.18%** in the OECD comparison group. Mean ESCS is also much lower in Rwanda (**-2.15**) than Mongolia (**-0.78**) or the OECD comparison group (**-0.19**).

Only **9.36%** of Rwanda's represented students attend schools in the two most urban location categories, compared with **47.06%** in Mongolia and **46.24%** in the OECD comparison group. Rwanda also reports higher educational material shortages (**0.67**) than Mongolia (**0.04**) and the OECD comparison group (**-0.07**). Staff-shortage means are **0.26** in Rwanda, **0.06** in Mongolia, and **0.28** in the OECD comparison group. These descriptive differences motivate examining whether mathematics performance varies with repetition, ESCS, and school-resource conditions.

8. Association with Mathematics performance

8A. Merge Student and School Data

[Show code](#)

8B. Weighted Mathematics Performance by Category

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8C. Gender and Grade Repetition

=== Mathematics Performance by Gender and Grade Repetition ===							
Analysis Group	Female Math	Male Math	Gender Gap (Male - Female)	No Repetition	Repeated Grade	Repetition Gap (Repeated - No Repetition)	
Mongolia	422.75	414.38	-8.37	424.08	332.92	-91.17	
Rwanda	301.73	324.70	22.97	339.25	293.39	-45.87	

8D. Mathematics Performance Across ESCS Ranges

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Next steps: [Generate code with escs_range_counts](#) [New interactive sheet](#)

8E. ESCS Missingness Check

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8F. Educational Material Shortages

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8G. Mathematics Performance by School Location

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=== Mathematics Performance by School Location ===

	Mongolia	Rwanda	OECD	
Village/Rural (<3,000)	376.00	304.00	410.15	
Small Town (3,000-15,000)	377.95	317.93	439.82	
Town (15,000-100,000)	413.14	308.84	466.71	
City (100,000-1M)	NaN	333.47	463.48	
Large City (1M-10M)	438.09	358.35	464.36	
Megacity (>10M)	NaN	349.19	480.11	

8H. Weighted Correlation Analysis

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=== Weighted Correlations with Mathematics Performance ===

	ESCS	Material Shortage	Staff Shortage	
Analysis Group				
Mongolia	0.376	-0.124	-0.017	
Rwanda	0.143	-0.161	-0.098	
OECD	0.395	-0.103	-0.049	

8I. ESCS and Mathematics Visualization

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Analysis Note: Associations with Mathematics Performance

Gender differences are modest relative to other patterns. Female students score **8.37 points higher** than males in Mongolia, while males score **22.97 points higher** in Rwanda and **18.70 points higher** in the OECD comparison group.

Grade repetition shows a much larger association. Students who repeated a grade score **91.17 points lower** than non-repeaters in Mongolia, **45.87 points lower** in Rwanda, and **85.19 points lower** in the OECD comparison group. These differences are descriptive and do not show that repetition causes lower achievement.

Mathematics generally rises with socioeconomic status. Mongolia increases from **373.89** below ESCS -2.0 to **484.69** above 1.0 , while the OECD comparison group rises from **381.73** to **516.69**. Rwanda rises from **308.22** below -2.0 to **356.10** between 0 and 1 before falling to **348.51** above 1.0 ; only **36 Rwandan students** are in that highest category, so the decline should not be overinterpreted. Mongolia remains above Rwanda within every common ESCS range. Weighted ESCS correlations with mathematics are **0.376** in Mongolia, **0.143** in Rwanda, and **0.395** in the OECD comparison group.

Students with missing ESCS score lower than students with observed ESCS by about **70 points in Mongolia, 48 in Rwanda, and 24 in the OECD comparison group**, reinforcing the missing-data limitation.

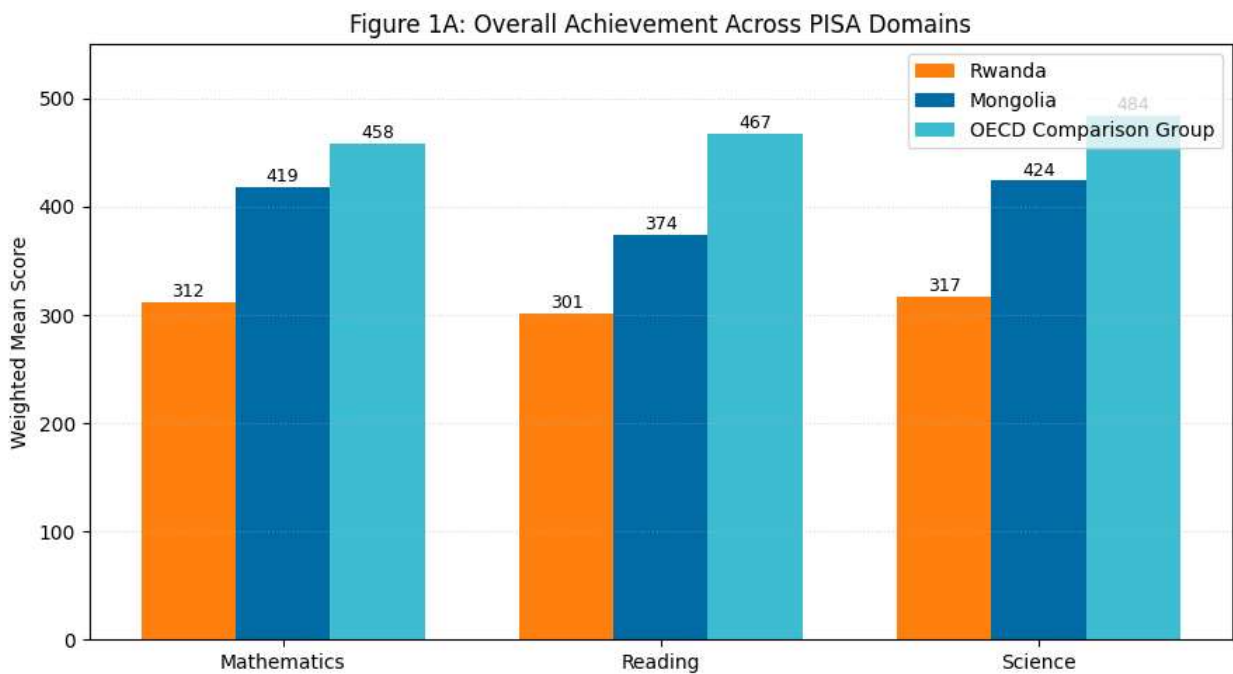
Educational material shortages are negatively associated with mathematics in all three groups. Scores fall from low- to high-shortage categories by **22.66 points in Mongolia, 30.21 in Rwanda, and 23.19 in the OECD comparison group**. Weighted correlations are **-0.124 , -0.161 , and -0.103** , respectively. Staff-shortage correlations are weaker: **-0.017 , -0.098 , and -0.049** . Overall, ESCS and grade repetition show the clearest associations examined here; none of these descriptive relationships establishes causality.

Policy Implications

For Rwanda, the results point to two priorities for further policy evaluation: educational material constraints and the mechanisms behind the unusually high rate of grade repetition, particularly among socioeconomically disadvantaged students. Because this analysis is observational, these patterns should guide further investigation and program evaluation rather than be interpreted as evidence that changing any single factor would cause higher mathematics performance.

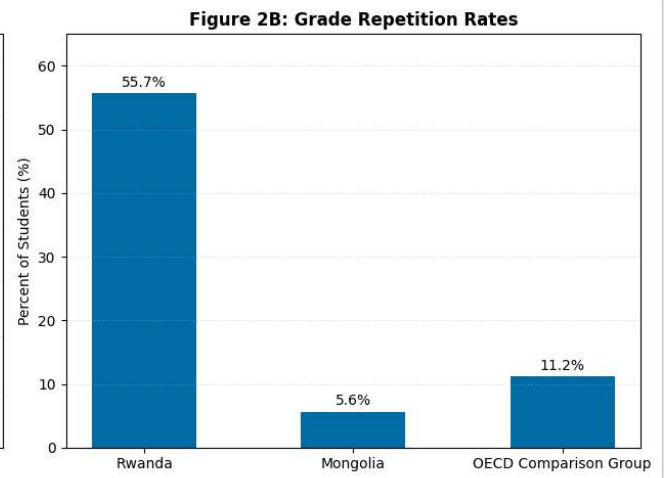
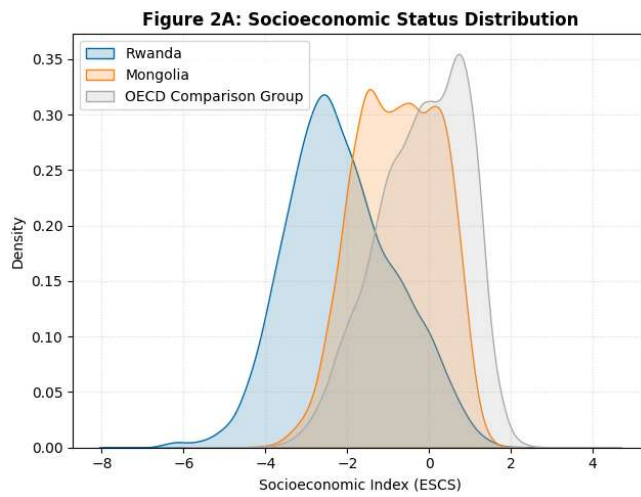
- ✓ 9. Outcome visualization
- ✓ 9A. Overall PISA Performance

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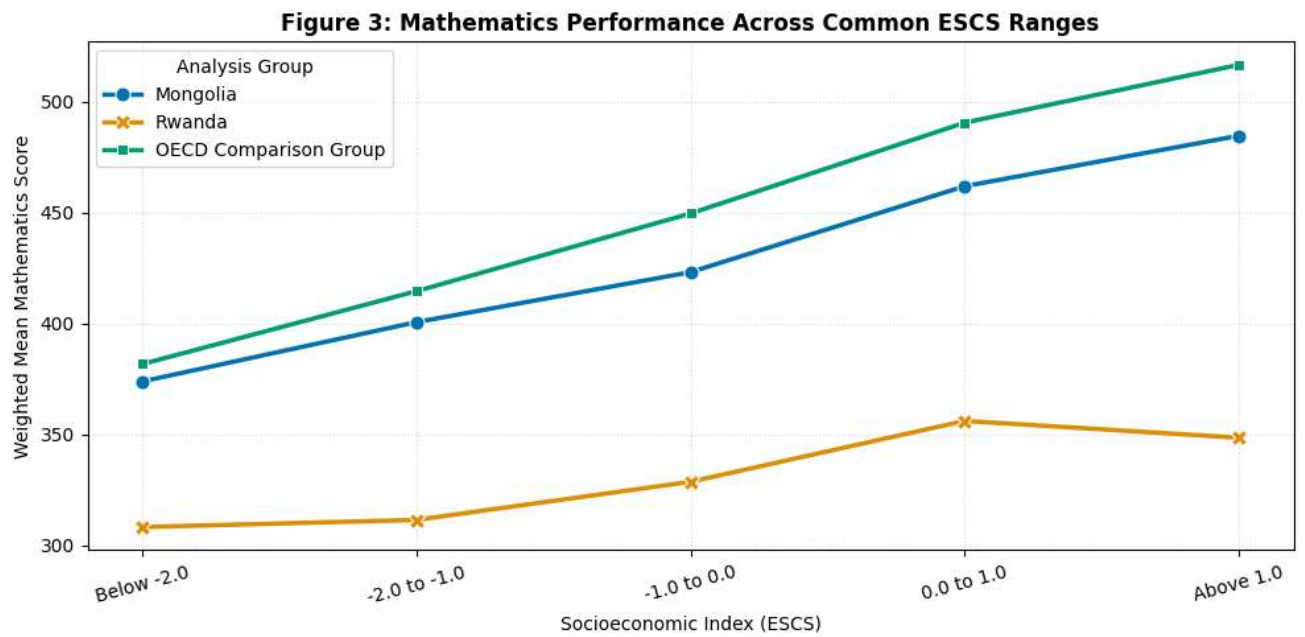
- ✓ 9B. Student Population Characteristics

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9C. Mathematics Across ESCS Ranges

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9D. Gender and Material Shortage Comparisons

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Figure 4A: Mathematics Performance by Gender

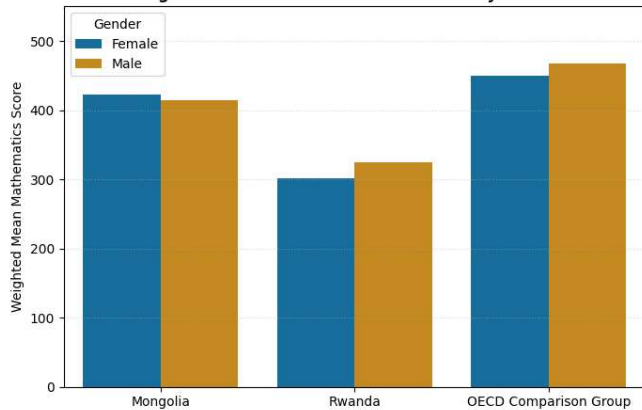


Figure 4B: Mathematics Performance by Material Shortage

